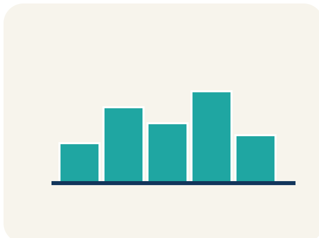
Shape of Data Distributions

Lesson 8-7

Name: ____________________ Date: ____________________ Class: ____________________

Key Vocabulary (Level 2 Standard)

Picture first, then the word, then a plain-language meaning. Say each word out loud.



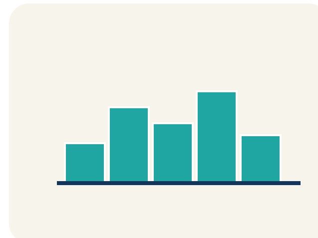
Heights: 60, 62, 64, 66, 68, 66, 64, 62, 60 — rises and falls evenly like a hill

Symmetric

Write the definition:

___

___



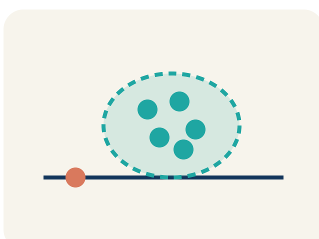
Skewed right: most data low (1,2,2,3,3,3,15) — tail stretches toward the high value

Skewed

Write the definition:

___

___



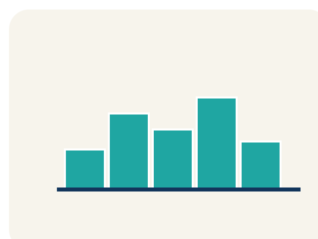
Scores 18, 19, 20, 21, 22 form a cluster around 20 — they are tightly grouped

Cluster

Write the definition:

___

___



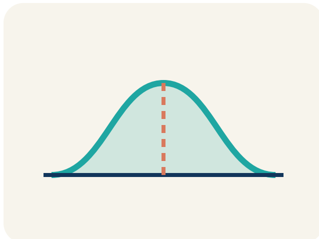
Data: 5, 6, 7, 8, ____, ____, ____, 20 — the empty space from 9 to 19 is a gap

Gap

Write the definition:

___

___



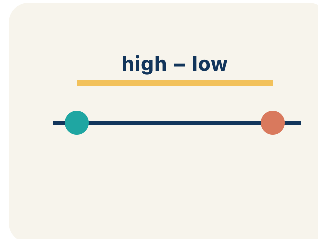
Looking at a dot plot or histogram, you can see if data is symmetric, skewed, clustered, or has gaps

Data distribution

Write the definition:

____

____



Symmetric data with low variability: tightly packed around the center. Skewed data: spread unevenly

Variability

Write the definition:

____

____

Guided Notes

Level 2 Standard

WHAT WE'RE LEARNING TODAY

I can describe the shape of a data distribution as symmetric, skewed, or having clusters and gaps.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Symmetric | **Skewed** | **Cluster** | **Gap** | **Data distribution** | **Variability**

1. A distribution where both sides look like mirror images is ______________ .
2. A distribution with a tail stretching out to one side is ______________ .
3. A group of data values bunched close together is a ______________ .
4. An interval where there are no data values is a ______________ .
5. The way data values are spread out across a range is the ______________ .
6. How much the values in a data set differ from one another is the ______________ .

Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 **I do — watch**

Follow each step as your teacher solves it.

Problem: A histogram of basketball players' ages shows: most players are 22–28, with a long tail of older players up to age 40. What is the shape of this distribution?

A. Skewed right

B. Symmetric

C. Skewed left

D. Uniform

Step 1 Most data is on the left (younger ages) with a tail stretching right (older ages).

Step 2 This is skewed right.

✅ **Answer:** A. Skewed right

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: A data set is symmetric. What is likely true about the mean and median?

A. They are approximately equal

B. The mean is much higher than the median

C. The median is much higher than the mean

D. They are always exactly the same

Step 1 ______________________________

Step 2 ______________________________

Answer: ______________________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Match each description with the correct distribution shape.

Show your work:

____

____

____

Try It

Solve on your own. Check the answer key when you are done.

1. Display 2: A dot plot of starting players' heights has one clear peak in the middle, with roughly even sides falling away from that peak. What shape best describes this distribution?

A. Symmetric with one cluster around the peak

B. Skewed right

C. Skewed left

D. Has a large gap

Show your work:

2. Display 5: A free-throw percentage histogram has most players clustered between 70% and 85%, a clear empty interval from 50% to 65%, and a few players down at 40%-50%. Which features are present, and what is the overall shape?

A. A cluster (70-85%), a gap (50-65%), and a left tail — so skewed left

B. A symmetric display with no gaps

C. A right tail of high values — so skewed right

D. Two equal peaks with no cluster or gap

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A teacher shows two dot plots of quiz scores from two different classes. Class A's data is symmetric with a cluster around 80. Class B's data is skewed right with most scores between 60–70 and a few scores near 100. Compare the classes: which has a higher median? Which has a higher mean? In which class does the mean better represent a typical score?

Sentence starter: Class A is ____ with scores clustered around ____. Class B is skewed ____ with most scores at ____. Class A likely has a ____ median. Class B's mean is pulled ____ by the high scores. The mean better represents a typical score in Class ____ because ____.

Show your work:

Reflect — Exit Ticket

A data set has most values clustered between 40–60, with a few values at 90–100. What is the shape of the distribution and which measure of center is best?

A. Skewed right; use median

B. Symmetric; use mean

C. Skewed left; use median

D. Uniform; use mean

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Symmetric
2. **Notes 2:** Skewed
3. **Notes 3:** Cluster
4. **Notes 4:** Gap
5. **Notes 5:** Data distribution
6. **Notes 6:** Variability
7. **We Try (notes):** A. They are approximately equal *— In a symmetric distribution, the mean and median are approximately equal because the data is balanced on both sides.*
8. **Try It 1:** A. Symmetric with one cluster around the peak *— One peak in the middle with roughly even sides means the data mirrors around the center, so it is symmetric with a single cluster at the peak.*
9. **Try It 2:** A. A cluster (70-85%), a gap (50-65%), and a left tail — so skewed left *— The data clusters at the high end (70-85%), there is an empty gap (50-65%), and the few low values form a tail on the left, so the display is skewed left.*
10. **Exit Ticket:** A. Skewed right; use median *— The data clusters on the left (40–60) with a tail to the right (90–100). This is skewed right. The median is best because the mean would be pulled toward the high values.*